

Alkylphenols and alkylphenol ethoxylates contamination of crustaceans and fishes from the Adriatic Sea (Italy).

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This paper presents the results of an investigation on the occurrence of alkylphenols (APs) and their ethoxylates (APEs) in 8 edible marine species from the Adriatic Sea and tries to estimate the corresponding intake for the Italian population. Two crustaceans, *Nephrops norvegicus* (Norway lobster) and *Squilla mantis* (spottail mantis shrimp), plus six fish species, *Engraulis encrasicolus* (anchovy), *Scomber scombrus* (Atlantic mackerel), *Merluccius merluccius* (European hake), *Mullus barbatus* (red mullet), *Solea vulgaris* (common sole) and *Lophius piscatorius* (angler) were analyzed for their content of nonylphenol (NP), octylphenol (OP) and octylphenol polyethoxylates (OPEs). These compounds were found in all analysed samples. NP was detected at the highest concentrations: 118-399 and 9.5-1431 ng g⁻¹ fresh weight (fw) respectively in crustaceans and fish. OP was found at respective levels of 2.7-4.7 and 0.3-3.8 ng g⁻¹ fw in crustaceans and fish, whereas OPE was determined at respective concentrations of 1.2-16.8 and 0.2-21.1 ng g⁻¹ fw in the same species. These results, together with those from a previous study on 4 edible mollusc, allow to estimate respective daily intakes for NP, OP, and OPE of about 12, 0.1, and 0.1 microg day⁻¹ for an Italian adult living along the Adriatic Coast. In relation to NP and OP, these intakes are much lower than the doses associated with toxic effects in laboratory animals (9 mg kg⁻¹ bw for rats). Nevertheless, data of exposure from other sources to these chemicals and others with similar biological characteristics are needed.